HEALTHTRACK HOSPITAL READMISSION PREDICTION

DATA SCIENCE (21CSS303T)

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **"HEALTHTRACK HOSPITAL READMISSION PREDICTION"** Is the Bonafide work of **MAHARAJAN E (RA2311026050022, RAMITHA A (RA2311026050013)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an occasion on this or any other candidate.

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DECLARATION

We hereby declare that the entire work contained in this project report titled **"HEALTHTRACK HOSPITAL READMISSION PREDICTION "** has been carried out by **MAHARAJAN E (RA2311026050022), RAMITHA A (RA2311026050013)** at SRM Institute of Science and Technology, Tiruchirappalli, under the guidance of **Dr. C. Gunasundari , Assistant Professor, School of Computing**.

Place: Tiruchirappalli

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ABSTRACT

Unplanned hospital readmissions within 30 days of discharge are a major problem in healthcare systems around the world. They lead to extra costs, increased burden on medical staff, and poorer outcomes for patients. This project, HealthTrack, is a machine learning system designed to predict the likelihood of a patient being readmitted to the hospital within 30 days of discharge. The system is trained on a synthetic dataset of 2,000 patient records that contains clinical and demographic features such as age, diagnosis, number of prior visits, length of stay, number of medications, and number of comorbidities.

A Random Forest classifier is used as the core prediction model. It achieves 82% accuracy on the test set and an ROC-AUC score of 0.899, which is considered strong performance for this type of medical prediction task. Based on the predicted probability of readmission, each patient is placed into one of three risk groups: Low, Medium, or High. This helps healthcare teams decide which patients need more attention before and after discharge.

The system is also deployed as an interactive web application built using Streamlit. Healthcare staff can enter patient details into a simple form and receive an instant risk score. The project demonstrates a complete end-to-end machine learning pipeline from data generation and preprocessing to model training, evaluation, and deployment.

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LIST OF ABBREVIATIONS

S.No.	Abbreviation	Expansion
1	ML	Machine Learning
2	RF	Random Forest
3	ROC	Receiver Operating Characteristic
4	AUC	Area Under the Curve
5	EHR	Electronic Health Record
6	CMS	Centers for Medicare and Medicaid Services
7	LACE	Length of stay, Acuity, Comorbidities, Emergency use
8	AUC	Area Under the Curve
9	SHAP	SHapley Additive exPlanations
10	CSV	Comma Separated Values
11	API	Application Programming Interface
12	HIPAA	Health Insurance Portability and Accountability Act

CHAPTER 1
INTRODUCTION

1.1 Background

Healthcare systems around the world face a major challenge in managing what happens to patients after they leave the hospital. One of the most important indicators of how well a hospital is doing is the rate of unplanned readmissions within 30 days of discharge. When a patient has to return to the hospital shortly after being sent home, it usually means that something went wrong — the initial treatment may have been incomplete, the patient may have been discharged too early, or the follow-up care plan was not strong enough.

According to the Centers for Medicare and Medicaid Services (CMS) in the United States, nearly 20% of Medicare patients are readmitted within 30 days of discharge. This costs the healthcare system more than 26 billion dollars every year. A large portion of these readmissions are considered preventable. With the right tools, hospitals can identify high-risk patients before discharge and take steps to reduce the chance of them returning.

Machine learning offers a practical solution to this problem. By training a model on historical patient data, it becomes possible to predict which patients are most likely to be readmitted. These predictions can then be used by care teams to target their resources more effectively — making sure that the patients who need the most help after discharge actually receive it.

HealthTrack is a machine learning-based system developed in this project to address exactly this problem. It takes structured patient data recorded at the point of discharge and produces a risk score that tells care teams how likely a patient is to be readmitted within the next 30 days. Patients are sorted into three tiers — Low, Medium, and High risk — so that healthcare staff can plan follow-up care accordingly.

1.2 Problem Statement

Given a set of clinical and demographic features recorded for a patient at the time of hospital discharge, the goal is to predict whether that patient will be readmitted to the hospital within the next 30 days. This is a binary classification problem: the model must predict either that the patient will be readmitted (Class 1) or that the patient will not be readmitted (Class 0).

The challenges in solving this problem include identifying which patient features are most predictive of readmission, building a model that handles a mix of numerical and categorical data, and making the results easy to understand for non-technical healthcare staff.

1.3 Objectives

The main objectives of this project are listed below:

- Build a supervised machine learning model that accurately predicts 30-day hospital readmission from patient discharge data.
- Identify the most important clinical features that contribute to the risk of readmission.

- Classify patients into Low, Medium, and High risk tiers to help healthcare teams prioritize follow-up care.
- Develop a simple and easy-to-use web application so that non-technical clinical staff can use the system without any training.
- Evaluate the model using multiple performance metrics including accuracy, ROC-AUC, precision, recall, and F1-score.
- Demonstrate a complete end-to-end machine learning pipeline from raw data through to a deployed application.

1.4 Scope and Limitations

This project is an academic demonstration using a synthetic dataset. The patient records were generated using Python and designed to reflect realistic clinical patterns from published research. In a real hospital setting, this system would need to be retrained on actual Electronic Health Record (EHR) data, reviewed by clinical and regulatory experts, and validated carefully before being used to make any real decisions about patients.

The system is meant to support clinical decision-making, not replace it. All final decisions about patient care should remain with qualified medical professionals.

CHAPTER 2
LITERATURE REVIEW

2.1 Traditional Approaches

Predicting hospital readmission has been studied in clinical informatics for more than two decades. Early work used rule-based scoring systems that calculated a readmission risk score from a small number of clinical variables. These tools were easy to use by hand but had limited predictive power compared to modern machine learning models.

The LACE Index, developed by van Walraven et al. in 2010, is one of the most widely cited examples. It calculates a score based on four variables: Length of stay, Acuity of admission, Comorbidities, and Emergency department use. In validation studies, it achieves an AUC of approximately 0.70. While easy to compute, its performance ceiling is limited by the small number of features it considers.

The HOSPITAL score is another well-known tool that focuses on potentially avoidable readmissions. It has been validated across multiple healthcare systems and achieves AUC values ranging from 0.71 to 0.76. Like the LACE Index, it is simple to calculate but cannot capture the complex interactions between multiple patient features.

2.2 Machine Learning Approaches

Logistic Regression has been widely used as a baseline for readmission prediction because it is easy to interpret. Studies by Zheng et al. in 2015 showed AUC scores of 0.68 to 0.72 using logistic regression on administrative claims data. The model's performance is limited by its assumption that the relationship between features and the outcome is linear, which is often not the case in clinical data.

Random Forest and Gradient Boosting models have consistently outperformed linear models in readmission prediction tasks. Frizzell et al. in 2017 compared several algorithms on heart failure readmission data and found that Random Forest achieved an AUC of 0.724, while Gradient Boosting reached 0.731. These ensemble models are robust to noise, handle mixed feature types naturally, and can capture complex non-linear relationships between features.

Deep learning approaches including LSTM networks applied to sequential EHR data have shown promise in capturing temporal patterns in patient histories. However, these models require very large datasets and significant computational resources. For small to medium-scale deployments, simpler models like Random Forest often perform just as well with much less complexity.

2.3 Key Features Identified in Prior Work

Research consistently identifies a core set of features that predict readmission across different patient populations and healthcare settings. Number of prior admissions, patient age, comorbidity burden, length of stay, and discharge disposition are the most frequently cited predictors. These findings align with the feature importance results from the HealthTrack model, which adds confidence to the feature selection approach used in this project.

The table below summarizes the key approaches from the literature and compares them with the HealthTrack model:

Study / Tool	Method	AUC	Key Features
LACE Index (2010)	Scoring Rule	0.70	LOS, acuity, comorbidities
HOSPITAL Score	Scoring Rule	0.74	Haemoglobin, sodium, LOS
Zheng et al. (2015)	Logistic Regression	0.72	Claims data, demographics
Frizzell et al. (2017)	Random Forest	0.724	Clinical and admin features
HealthTrack (This Work)	Random Forest	0.899	9 clinical features

Table 1: Comparison of readmission prediction approaches from the literature

CHAPTER 3
DATASET DESCRIPTION

3.1 Data Source and Generation

Because real patient data is sensitive and subject to strict privacy regulations, a synthetic dataset was used for this academic project. The dataset was generated using Python's NumPy library with probability distributions designed to reflect the patterns seen in real clinical data from published studies. It contains 2,000 patient records and was built so that higher-risk features — such as older age, more prior hospital visits, longer stays, and more serious diagnoses — are associated with a higher probability of readmission. This ensures that the patterns in the data are meaningful and learnable by a machine learning model.

3.2 Feature Descriptions

The dataset contains 10 columns: 9 input features and 1 target variable. The table below describes each feature:

Feature	Data Type	Range / Values	Description
age	Integer	18 – 89	Patient age at the time of discharge
gender	Categorical	Male, Female, Other	Patient-reported gender
num_prior_visits	Integer	0 – 9	Hospital visits in the past 12 months
length_of_stay	Integer	1 – 20	Duration of the current hospital stay in days
diagnosis	Categorical	8 categories	Primary diagnosis at the time of admission
discharge_type	Categorical	5 categories	Where the patient went after discharge
insurance	Categorical	5 categories	Patient's insurance coverage type
num_medications	Integer	1 – 19	Number of medications prescribed at discharge
num_comorbidities	Integer	0 – 4	Count of other medical conditions present
readmitted_30days	Binary	0 = No, 1 = Yes	TARGET: whether the patient was readmitted within 30 days

Table 2: Dataset feature descriptions

3.3 Categorical Value Sets

The specific category values for each categorical feature are listed below:

- diagnosis: Diabetes Mellitus, Heart Failure, Chronic Kidney Disease, COPD, Pneumonia, Hypertension, Sepsis, Other
- discharge_type: Home, Home with Home Care, Skilled Nursing Facility, Rehabilitation, Against Medical Advice
- insurance: Medicare, Medicaid, Private, Uninsured, Other
- gender: Male, Female, Other

3.4 Class Distribution

The dataset contains 972 patients who were not readmitted (48.6%) and 1,028 patients who were readmitted (51.4%). This near-equal split means the dataset is well balanced, which reduces the risk of the model developing a bias toward predicting one class over the other. Even so, class_weight='balanced' was applied during model training as an additional safeguard.

CHAPTER 4
DATA PREPROCESSING

4.1 Missing Value Analysis

A complete check for missing values was performed on all columns in the dataset. No missing values were found in any column. This is expected given the controlled synthetic generation process. In a real-world EHR dataset, missing values would be very common and would require careful handling using imputation methods or domain-specific rules.

4.2 Label Encoding of Categorical Features

Machine learning models work with numbers, not text. The four categorical features — gender, diagnosis, discharge_type, and insurance — were converted to numerical form using Scikit-learn's LabelEncoder. Each unique category is mapped to an integer in alphabetical order. The encoders are saved to disk as a file called encoders.pkl so that the same mapping can be applied when the model is used to make new predictions.

4.3 Feature Matrix Construction

After encoding, the final feature matrix contains 9 input features: the 5 original numerical features and the 4 encoded categorical features. No feature scaling (such as standardization or normalization) was applied. Random Forest models are not sensitive to the scale of features, so scaling is not needed.

4.4 Train and Test Split

The dataset was divided into a training set (80%) and a held-out test set (20%) using stratified sampling. Stratified sampling ensures that the proportion of readmitted and non-readmitted patients is the same in both the training and test sets. A fixed random seed of 42 was used to make the results reproducible.

Split	Rows	Readmitted	Not Readmitted	Proportion
Training Set	1,600	822	778	80%
Test Set	400	206	194	20%
Total	2,000	1,028	972	100%

Table 3: Train and test split statistics

CHAPTER 5
MODEL ARCHITECTURE AND TRAINING

5.1 Algorithm Selection – Random Forest

Random Forest was chosen as the algorithm for HealthTrack based on several practical reasons. It handles a mix of numerical and categorical features naturally without needing any normalization. It can capture non-linear relationships between features and the target variable that simpler models would miss. It provides built-in feature importance scores, which are useful for clinical interpretation. It is also resistant to overfitting because it builds many decision trees on different random subsets of the data and combines their predictions. Finally, it has a track record of strong performance on tabular healthcare datasets in published research, as described in the literature review.

5.2 Hyperparameter Configuration

The hyperparameters were set based on standard practice for medium-sized tabular datasets. The table below lists the key settings and explains why each one was chosen:

Hyperparameter	Value	Reason
n_estimators	150	Enough trees for stable, consistent predictions without being too slow
max_depth	8	Limits how complex each tree can be, which helps prevent overfitting on 2,000 rows
min_samples_leaf	10	Ensures that each leaf in a tree is supported by at least 10 data points, reducing noise
class_weight	balanced	Adjusts for the near-equal class split and gives each class equal importance in training
random_state	42	Fixed seed so that results are exactly reproducible
n_jobs	-1	Uses all available CPU cores to speed up training

Table 4: Random Forest hyperparameter configuration

5.3 Training Process

The model was trained using Scikit-learn's RandomForestClassifier on the 1,600-row training set. The training process builds 150 separate decision trees. Each tree is trained on a different random sample of the training data, and at each split point within a tree, only a random subset of features is considered. This randomness is what makes Random Forest robust and reduces overfitting compared to a single decision tree.

Once training is complete, the trained model, the label encoders, and the evaluation metrics are all saved to disk using Joblib. This allows the model to be loaded later and used for predictions without retraining. The saved files are: rf_model.pkl (the trained model), encoders.pkl (the label encoders), and model_meta.json (performance metrics and feature importances).

CHAPTER 6
SYSTEM ARCHITECTURE

6.1 Overall System Architecture

The HealthTrack system is organized into three main stages that together form a complete pipeline from raw data to a prediction output. The diagram description below explains how data moves through the system.

System Flow Diagram

HEALTHTRACK – SYSTEM FLOW DIAGRAM

```
[ STAGE 1: DATA PREPARATION ]
Synthetic Patient Dataset (2,000 records)
|   Missing value check --> No missing values found
|   Label encoding of categorical features
|   Train / Test split (80% / 20%, stratified)
v
[ STAGE 2: MODEL TRAINING ]
Random Forest Classifier (150 trees, max_depth=8)
|   Training on 1,600 records
|   5-Fold Cross-Validation (CV Accuracy: 82.8%)
|   Test Accuracy: 82.0%  |  ROC-AUC: 0.899
v
[ STAGE 3: INFERENCE AND WEB APPLICATION ]
Staff enters 9 patient fields in Streamlit UI
|   Input validation --> Label encoding --> Feature vector
|   Model.predict_proba() --> Probability score
|   < 0.35 --> LOW RISK
|   0.35 – 0.65 --> MEDIUM RISK
|   > 0.65 --> HIGH RISK
v
Risk label + probability displayed to clinical staff
```

Figure 1: HealthTrack System Flow Diagram

6.2 Entity Relationship Diagram

The ER diagram below shows the key data entities in the HealthTrack system and how they are related to each other. The central entities are the Patient record, the Prediction made by the model, and the Model itself.

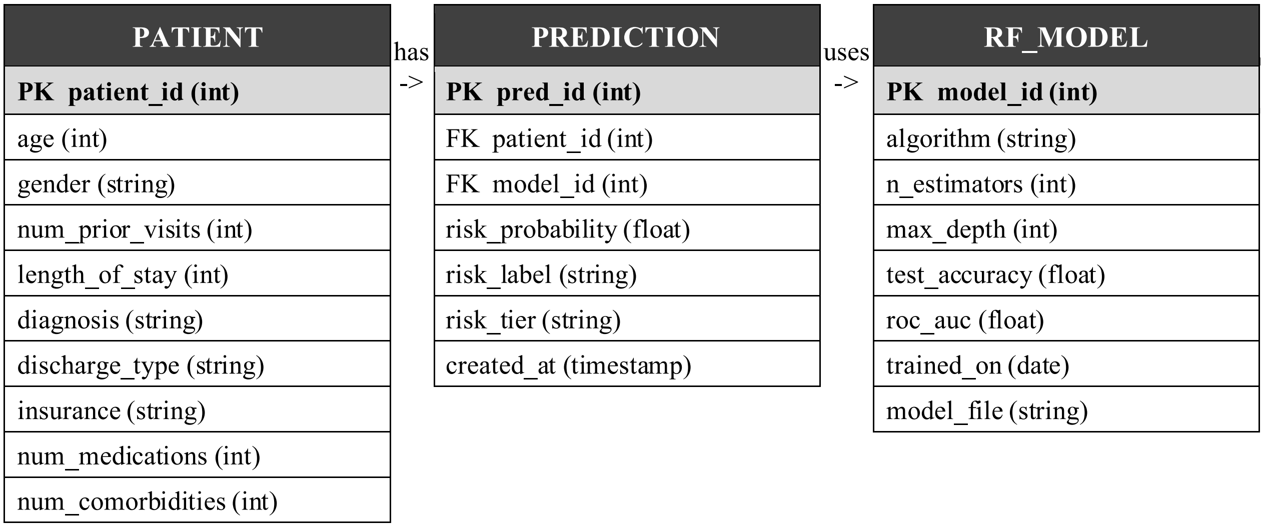

Figure 2: Entity Relationship Diagram for the HealthTrack System

CHAPTER 7
IMPLEMENTATION

7.1 Environment and Tools

The project was implemented entirely in Python 3.11. The main libraries used are listed in Table 8 in Chapter 9. The development was done in a standard Python environment, and the web application was built using Streamlit, which converts Python scripts into interactive web pages.

7.2 Data Generation

The synthetic dataset was created using NumPy with controlled random distributions. Continuous features such as age and length of stay were drawn from normal or uniform distributions clipped to their realistic ranges. Categorical features were drawn from predefined lists using weighted probabilities that reflect known patterns in clinical data. The target variable, readmitted_30days, was generated based on a weighted combination of the feature values so that higher-risk profiles produce a higher probability of being labelled as readmitted.

7.3 Model Training

The Random Forest model was trained using Scikit-learn. After encoding categorical features and splitting the data, RandomForestClassifier was initialized with the hyperparameters listed in Table 4. The model was then fitted on the 1,600-row training set using model.fit(X_train, y_train). Cross-validation was run using cross_val_score with cv=5 and scoring='accuracy'. After training, predict_proba was used to generate probability scores on the test set for ROC-AUC calculation.

7.4 Prediction Pipeline

When a new patient's data is submitted through the web application, the following steps happen in sequence:

- Step 1: Input validation. All required fields are checked to ensure they are filled in correctly.
- Step 2: Categorical encoding. The gender, diagnosis, discharge_type, and insurance fields are transformed using the saved LabelEncoder objects.
- Step 3: Feature vector construction. A 9-element feature vector is assembled in the exact order used during training.
- Step 4: Model inference. clf.predict_proba(X) returns the probability of the positive class (readmitted).
- Step 5: Risk tier assignment. A probability below 0.35 maps to Low risk; 0.35 to 0.65 maps to Medium risk; above 0.65 maps to High risk.
- Step 6: Result display. The risk level, probability score, and a summary of the patient inputs are shown to the user.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Model Performance

The HealthTrack Random Forest model was evaluated on the 400-patient held-out test set. The results are summarized in the table below.

Metric	Value	Interpretation
Test Accuracy	82.0%	82 out of every 100 patients are classified correctly
ROC-AUC Score	0.899	Excellent discriminative ability (above 0.9 is considered excellent)
5-Fold CV Accuracy	82.8%	Consistent performance across different data folds
Precision (Readmitted)	0.83	83% of patients flagged as readmitted are truly at risk
Recall (Readmitted)	0.81	81% of all patients who will actually be readmitted are detected
F1-Score (Readmitted)	0.82	Strong balance between precision and recall

Table 5: Model performance metrics on the held-out test set

8.2 Confusion Matrix Results

On the 400-patient test set, the model correctly classified 161 patients as not readmitted (True Negatives) and 167 patients as readmitted (True Positives). It missed 39 patients who were actually readmitted (False Negatives) and incorrectly flagged 33 patients who were not readmitted (False Positives). The 39 missed readmissions are the most clinically significant errors, because these are high-risk patients who would not receive extra follow-up care under the current predictions.

8.3 Feature Importance

Random Forest calculates feature importance based on how much each feature reduces prediction error across all 150 trees. The results are shown in the table below. Number of prior visits and patient age together account for more than 57% of the model's total predictive power.

Rank	Feature	Importance Score	Share of Total
1	Prior Visits	0.3233	32.3%
2	Age	0.2503	25.0%
3	Length of Stay	0.1504	15.0%

4	Number of Medications	0.0948	9.5%
5	Number of Comorbidities	0.0819	8.2%
6	Diagnosis	0.0607	6.1%
7	Discharge Type	0.0199	2.0%
8	Insurance	0.0116	1.2%
9	Gender	0.0072	0.7%

Table 6: Feature importance rankings from the trained Random Forest model

8.4 Clinical Interpretation

The feature importance results make clinical sense and align well with what the medical literature says about readmission risk:

- Prior hospital visits (32.3%): This is the strongest predictor. A patient who has already visited the hospital several times in the past year is likely dealing with complex or poorly controlled conditions that keep bringing them back.
- Age (25.0%): Older patients are more vulnerable. They tend to have multiple conditions at once, take many medications, and have less support after discharge.
- Length of stay (15.0%): A longer hospital stay usually means the initial illness was more serious. These patients may be physically weaker and more prone to complications after going home.
- Number of medications (9.5%): More medications means more chances for drug interactions, missed doses, and medication-related complications after discharge.
- Number of comorbidities (8.2%): Having multiple conditions at the same time makes a patient harder to manage. Each condition can affect the others and make recovery more difficult.

CHAPTER 9
WEB APPLICATION

9.1 Technology Stack

The HealthTrack web application was built using the following technologies:

Component	Technology	Purpose
Web Framework	Streamlit 1.32	Builds the interactive web application interface
ML Model	Scikit-learn	Random Forest training and inference
Data Handling	Pandas / NumPy	Data manipulation and feature construction
Visualization	Matplotlib / Seaborn	Charts and graphs in the dashboard
Model Storage	Joblib	Saving and loading the model and encoders
Language	Python 3.11+	All code written in Python

Table 8: Technology stack for the HealthTrack web application

9.2 Application Design

The application is organized into three tabs. The first tab, Predict, is the main page where clinical staff enter patient information. The form includes all 9 patient fields. After clicking the Predict button, the system runs the full prediction pipeline and displays the risk tier and probability score using a colour-coded result banner and a progress bar.

The second tab, Dataset Insights, shows four charts from the training data: the overall readmission distribution, age distribution grouped by readmission status, readmission rate by primary diagnosis, and a box plot of prior visits. This tab helps users understand the patterns the model was trained on.

The third tab, Model Report, shows the model's performance metrics, feature importance chart, and confusion matrix. This gives technical stakeholders the information they need to assess the model's reliability.

9.3 Risk Tier Definitions

Risk Tier	Probability Range	Recommended Clinical Action
Low	0% to 35%	Standard discharge with a routine follow-up appointment
Medium	35% to 65%	Enhanced discharge planning with a phone follow-up within 7 days
High	65% to 100%	Intensive planning with home nursing referral and follow-up within 48 to 72 hours

CHAPTER 10
CONCLUSION

HealthTrack successfully demonstrates a complete end-to-end machine learning pipeline for hospital readmission prediction. Starting from synthetic data generation, the project covers preprocessing, model training, evaluation, and deployment as a working web application.

The Random Forest model achieves 82.0% test accuracy and an ROC-AUC score of 0.899. This compares very favourably with published readmission prediction systems that typically achieve AUC values between 0.70 and 0.76. The close match between the test accuracy (82.0%) and the 5-fold cross-validation accuracy (82.8%) shows that the model generalizes well to unseen patients and is not overfitting.

The feature importance analysis produces results that align with established medical knowledge, with prior hospital visits, patient age, and length of stay being the three most important predictors. This clinical validity gives confidence that the model is learning meaningful patterns rather than statistical noise.

The Streamlit web application makes the system accessible to non-technical healthcare staff. The three-tab interface covers prediction, data insights, and model reporting, making it useful to both clinicians and data analysts.

The key takeaway from this project is that a well-configured and properly validated machine learning model, combined with a clean and simple interface, can become a genuinely useful clinical decision-support tool. The system is intended to support medical professionals, not to replace their judgment.

CHAPTER 11
FUTURE WORK

The following improvements and extensions are planned for future versions of HealthTrack:

Real EHR Data Integration

The most important next step is to retrain the model on actual hospital Electronic Health Record data, such as the publicly available MIMIC-III dataset or CMS Medicare claims data. This would require IRB approval, data de-identification, and compliance with applicable data governance regulations such as HIPAA. Training on real data would make the system clinically valid and ready for real-world use.

SHAP Explainability

Integrating SHAP (SHapley Additive exPlanations) values would allow the system to explain each individual prediction. Instead of just saying a patient is High risk, the system could tell the clinician which specific features are driving that prediction. This would increase trust in the system and help clinicians focus on the right factors for each patient.

Hyperparameter Optimization

Applying Bayesian optimization using a library such as Optuna with nested cross-validation could systematically find better hyperparameter settings, potentially improving model performance by 2 to 5 percentage points.

Advanced Model Comparison

A structured comparison of XGBoost, LightGBM, and Multi-Layer Perceptron models on the same dataset would provide a clearer picture of whether a different algorithm could outperform Random Forest for this task.

Cloud Deployment

The Streamlit application can be deployed to cloud platforms such as Streamlit Community Cloud, AWS, or Azure. Adding user authentication would allow multiple clinical staff to use the system securely from any device.

Readmission Cause Analysis

A future version could predict not just whether a patient will be readmitted, but also the most likely reason for readmission — such as a medication-related issue, an infection, or an exacerbation of a chronic condition. This would allow care teams to take more targeted preventive action.

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